

Functional Requirements

1. User Authentication & Role Management

- a. Users shall be able register and log in using university credentials.
- b. The system shall enforce role-based access control for students, event organizers, and admins.

2. Event Browsing & Search

- a. Students shall be able to browse events categorized by type (academic, cultural, sports, etc.).
- b. The search functionality shall allow finding events based on keywords, date, or category.

3. Event Registration & Ticketing

- a. Students shall be able to register for free events or purchase tickets for paid events using Stripe.

4. Event Notifications

- a. Students shall receive email notifications for event registrations, updates, and reminders.
- b. Students shall get real-time notifications for last-minute changes or cancellations.

5. Volunteer Registration & Management

- a. Students shall be able to sign up for volunteering opportunities in events.
- b. Event organizers shall be able to set volunteer requirements and approve applicants.

6. Event Creation & Management

- a. Organizers shall be able to create and edit event details (date, location, capacity, ticket price).
- b. Organizers shall be able to view real-time event registration stats and ticket sales.

7. Post-Event Feedback & Reviews

- a. Attendees shall be able to leave ratings and reviews for events.
- b. Feedback data shall be accessible to organizers for future improvements.

8. Event Analytics & Reports

- a. Admins shall generate reports on attendance, volunteer participation, and event popularity.

9. Admin Management & Moderation

- a. Admins shall approve or reject event listings based on university guidelines.
- b. Admins shall suspend user accounts violating policies.

Non-Functional Requirements (Measurable)

1. Performance

- a. The system should handle at least 150 concurrent users without performance degradation.

2. Security

- a. All user authentications must be secured using Django Auth with password hashing.
- b. Payment transactions should be secured by the encryption protocols in Stripe.

3. Usability

- a. 85% of users should be able to complete event registration within 2 minutes.

4. Data Integrity

- a. Event registration data should be backed up every 24 hours to prevent loss.

5. Response Time

- a. The system must load the homepage within 5 seconds for 90% of users.

6. Browser Compatibility

- a. The web app should work smoothly on popular browsers like Google Chrome, Firefox, and Safari (latest versions).

7. Web Responsiveness

- a. The web app should be fully responsive and function correctly across devices and screen sizes, from half-screen to full-screen mode, ensuring a seamless user experience (1024 px – 1600 px).

8. Session Timeout

- a. User sessions should automatically log out after 30 minutes of inactivity to improve security.